# TRACE-RPG: Source-Bound Ontology Evidence and a Bounded Codex-Assisted Integration Case in an Interactive Game

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***Abstract*—TRACE-RPG combines external authorization, typed-error repair, unchanged fallback, revalidation, replay, and accounting; its read-only bilingual ontology view binds source declarations to public observations without mutation authority. One operator-assisted Codex dashboard case ($n$=1, independent repeats 0) reports `gpt-6.1-sol`, repaired geometry, retained failures, 62 checks, a Godot Web build, and real-input denial/disclosure/redisplay; for this dashboard case, provider identity, revision, seed, cost, request count, and inference latency remain unavailable. A fixed-game native pilot on one authored quest records 8 assigned rows (7 observed): 5 system-goal successes, one retained pre-response harness-serializer technical failure (executed `null`, not rerun), and 2 `gpt-6.1-sol` ontology response-timeout failures; 23 host invocations, 0 repairs, 0/16 unsafe committed transitions, 7 valid semantic-link replays, and 6 valid fresh-engine reexecutions (1 `NOT_RUN`; first row unobserved). The pilot carries a retained controller-source revision confound (row 1 under controller-v3, rows 2–8 under v4), and provider identity, revision, cost, and request count are unknown. It is exploratory, with one stateless intent proposer and deterministic planner/checker; it does not establish confirmatory questions A (model/stack consistency) or C (fresh-reference noninferiority), ontology efficacy, ranking, or player benefit. Traceability, whole-game, and repeated-cross-model studies are proposals. The native pilot used seed 20261024. Separately, a synthetic v5 development collection received responses for 20/20 assignments; 8/20 passed the strict gate, 7/20 reached the specified goal, and 10/20 failed schema parsing. These descriptive counts include all assignments and do not establish a causal feedback effect or model superiority.**

***Index Terms*—game AI, interactive narrative, symbolic validation, constrained generation, reproducibility**

## I. Introduction

GENERATED events may request absent items, unknown facts, forbidden disclosures, unauthorized effects, or regressions. Plausibility [1], [2] and grounding [3], [4] do not authorize mutation: **only TRACE-RPG's external deterministic gate authorizes mutation** (Fig. 1).

Our contribution is a scoped proposal-to-commit integration and its evidence-accounting design: repair information can change the candidate, but only revalidation against the same authored constraints can authorize a state change; observed checks and replay have explicitly separate evidential limits. C1–C5 organize these contracts and their evidence, rather than claiming new constituent algorithms or demonstrated model advantage.

Manuscript prepared for anonymous review.

TABLE I
CONTRIBUTIONS, PRIOR TOPICS, EVIDENCE LANES, AND INFERENCE CEILINGS (GENERATED FROM THE STRUCTURED MATRICES)

| ID | Contribution | Prior topics (refs) | Lanes | Inference ceiling |
|---|---|---|---|---|
| C1 | Trust-boundary contracts | T1, T2, T3 (17) | E1, ENG1 | encoded-field contract conformance |
| C2 | Validate–repair–commit controller | T1, T8 (14) | E1, ENG1 | mechanism behavior on frozen cases |
| C3 | Audit-linked evidence layer | T4, T7, T8 (6) | E1, ENG1 | integrity/replay on named fixtures |
| C4 | Assignment-complete harness | T1, T4, T6, T7 (20) | E1 | exact designed-case accounting |
| C5 | Counterexample-guided operator $\rho$ | T1, T8, T9 (8) | E1 | frozen offline repairability classes |

T1 = grounded game worlds; T2 = retrieval/memory agents; T3 = structured generation; T4 = environments/benchmarks; T5 = player experience; T6 = human/LLM evaluation; T7 = statistics/reproducibility; T8 = planning/runtime interposition; T9 = feedback repair. Lanes are defined in Table V. All rows are authored-fixture verified; none supports an efficacy claim.

Table I scopes **C1—contracts**, **C2—validation/repair**, **C3—audit links**, **C4—assignment accounting**, and **C5—guided repair** to their evidence and inference ceilings.

## II. Related Work and Research Gap

Grounding/memory/planning/role [3], [4], [5], [6] and player/dialogue work [7], [8], [9], [10] distinguish believability from admissibility; affect/engagement [11], [12], [13] remains future, unimplemented, and non-authoritative. The design question addressed here is how to keep mutation authority, repair information, and replay evidence distinct within a source-bound proposal-to-commit path; this is not a claim that prior work lacks these components.

Benchmarks span state, embodiment, content, process, social interaction, and long-horizon play [1], [2], [14], [15], [16], [17], [18], [19], [20]; NCP-Bench tests narrative commitment/fact consistency, not player outcomes [21]. Structural/schema, affordance, and rule methods [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33] motivate constraints, not superiority; retrieval/memory supplies context without commit authority [34], [35], [36].

Reproducibility/evaluation motivates frozen artifacts, retained failures, construct/rater/bias controls, and justified margins/random effects [37], [38], [39], [40], [41], [42], [43], [44], [45]. Replay/leakage checks support denominators, not efficacy [46], [47]. C1–C5 combine enforcement, state/transaction management, feedback repair, action interposition, preconditions/effects, abort, social/belief-aware planning, and replay

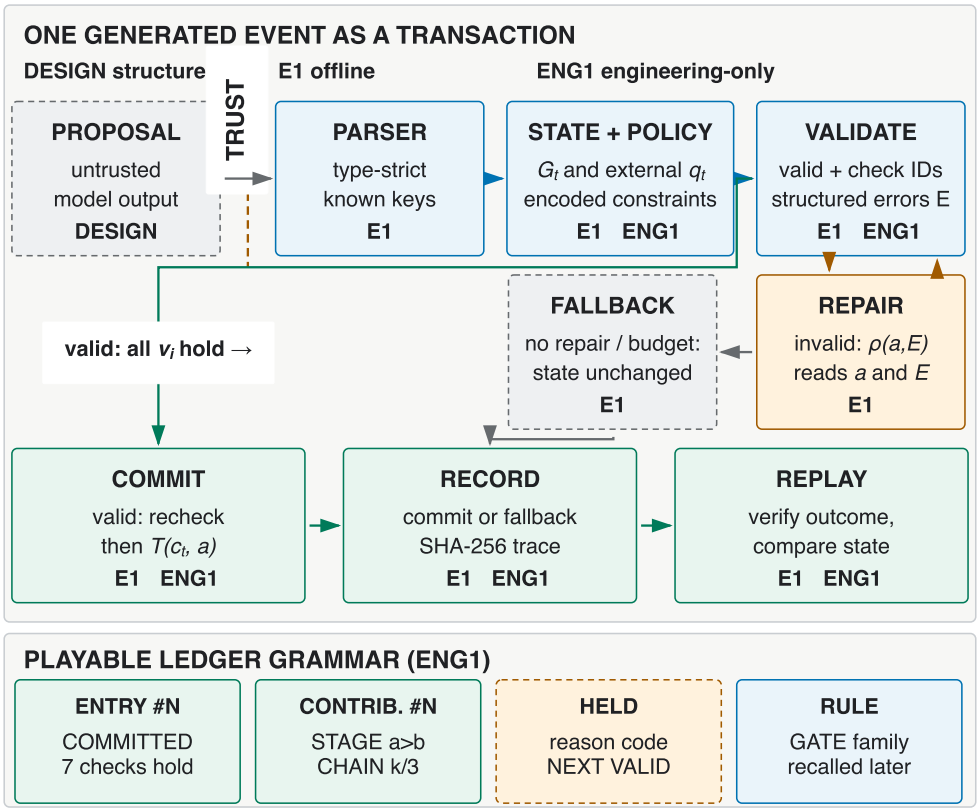



Fig. 1. Policy/$c_t$ gates proposals; errors feed bounded $\rho$ or unchanged fallback. Passing candidates are rechecked, committed, recorded, and replayed. Historical cohorts/legacy ENG1 remain archived, not active model evidence. Below: committed-snapshot ledger grammar.

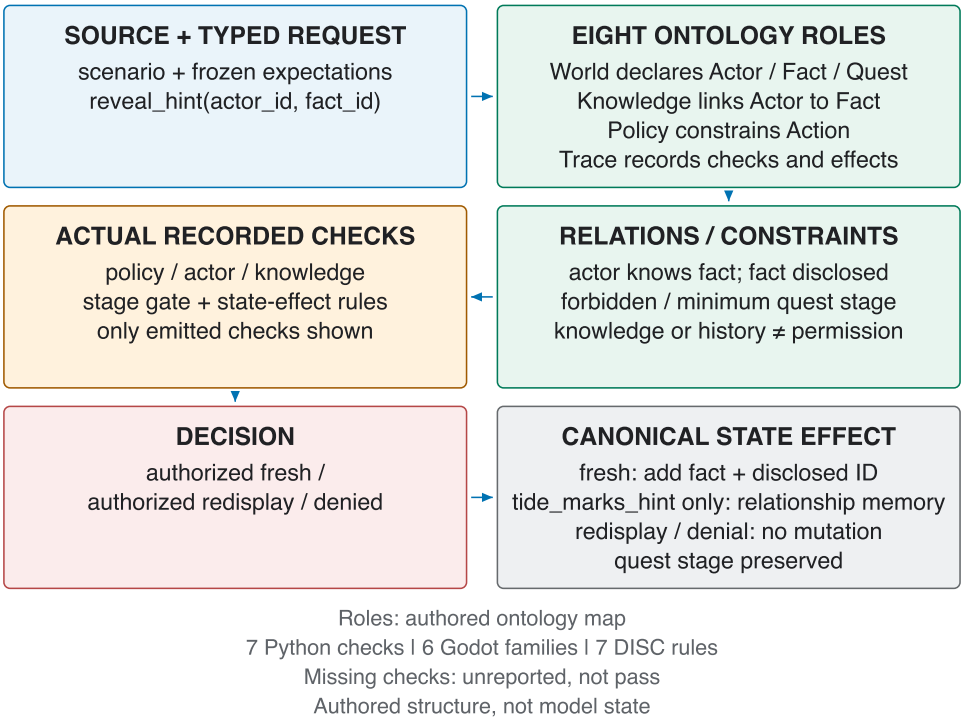



Fig. 2. Eight-role ontology structure (Table III) links source fields to checks and observed decisions. Only authorized transitions change canonical state; the diagram is neither live capture nor hidden-state evidence.

[27], [28], [29], [48], [46], [49], [50], [51], [52], [53], [54]; no invention/firstness claim.

## III. Problem Definition and Guarantee Boundary

$c_t = (G_t, q_t)$ separates the typed game snapshot $G_t$ from externally authored action-policy/disclosure/quest-stage constraints $q_t$. In Python these are carried together by `WorldState`: locations, reachable locations, object locations, inventory, facts, NPC knowledge, and quest stage describe the encoded state; action policies and forbidden disclosures constrain it. An `ActionPolicy` specifies required preconditions, allowed and required effects, and allowed quest-stage targets. This is a conceptual separation, not a claim that Python stores two independent objects. Predicates neither read $h_{<t}$ nor reconstruct `WorldState`; transitions update $G_t$ only, excluding policy rotation. Retrieval, scores, rationales, memory, and future affect lack authority.

After strict parsing of the untrusted proposal $a_t$, the following is a conceptual representation of validation, not the literal Python return type:

$$V(c_t, a_t) = (\boldsymbol{v}, E_t),$$
$$\boldsymbol{v} = (v_{\text{policy}}, v_{\text{pre}}, v_{\text{reach}}, v_{\text{know}}, v_{\text{disc}}, v_{\text{elig}}, v_{\text{stage}}) \in \{0,1\}^7, \tag{1}$$

Here $E_t$ denotes structured errors and the seven bits denote policy, precondition, reachability, NPC knowledge, forbidden disclosure, stage eligibility, and stage monotonicity, respectively. Python instead returns `ValidationResult(valid, errors, checks_run)`: a single validity flag, errors with code/entity/reason/repair hint, and seven check identifiers, not seven returned Boolean outcomes. A check identifier indicates the check was run, not that a separate per-check result was observed. An absent trace or omitted field is unobserved, not a passing check or a failed one. The following equation describes a single-current-candidate gate, not the whole transaction for the original proposal. Here $a_t$ is the current candidate, possibly changed by repair, and $\boldsymbol{v}$ denotes that candidate's checks against unchanged $c_t$. A rejected attempt leaves $c_t$ unchanged while control continues through bounded repair or fallback; only passing same-policy revalidation authorizes mutation:

$$c_{t+1} = \begin{cases} T(c_t, a_t), & \bigwedge_{i=1}^{7}(v_i = 1), \\ c_t, & \text{otherwise.} \end{cases} \tag{2}$$

The operational path is proposal → deterministic validation → observable validity, check identifiers and structured errors → optional bounded repair → revalidation against the same state → commit or unchanged fallback. A valid candidate is checked again immediately before application; exhausted repair or no repairer leaves the state unchanged. The guided repair operator described below uses only the proposal and errors; the general Python repair callback nevertheless receives `WorldState`, so the restricted-input claim does not apply to every possible callback. For a controlled comparison, deterministic and model-based detailed repair must receive the same proposal and detailed errors, while the same model's generic-error condition receives a generic signal; no comparison outcome is implied by this design.

TABLE II
Encoded Validator Boundary

| Predicate | Rejects | On failure |
|---|---|---|
| Action policy | unknown action/type; omitted required preconditions/effects; disallowed effects or unauthorized quest-stage effects | unchanged |
| Precondition | absent/false requirement | unchanged |
| Reachability | inaccessible object | unchanged |
| NPC knowledge | undeclared known fact | unchanged |
| Disclosure | forbidden fact | unchanged |
| Quest stage (`quest_stage`) | required stage above current | unchanged |
| Quest-stage monotonicity (`quest_stage_monotonicity`) | regressive stage | unchanged |

Fixtures test invariants, not theorems. **I1:** validation exhaustion or pre-commit proposal/controller failure preserves state. **I2:** returning callbacks and $K \in \mathbb{Z}_{\geq 0}$ bound attempts to $K + 1$ with pre-commit revalidation; without deadlines, wall-clock termination is not guaranteed. **I3:** accepted `CandidateAction.effects` stay within `ActionPolicy.allowed_effects`, including authorized quest-stage targets. **I4:** given the recorded initial snapshot, a complete ordered transition record, and the frozen schema and deterministic transition version, replay can reconstruct the recorded terminal structured state. This is conditional reconstruction, not authentication, semantic correctness, or reproduction of model generation.

Assumptions: correct policy/extraction, deterministic versions, returning callbacks, single-process ownership. Replay checks structured transitions, not repair provenance, prompts, stochastic generation, or undeclared natural-language implications.

## IV. TRACE-RPG Architecture and Algorithms

### A. Trust Boundary and Contracts

External models/adapters propose; parser rejects ambiguous types/unknown keys; independent policy defines preconditions/effects. Versioned bridge carries observations, proposals, validations, rejections, fallbacks, and commits (Fig. 1); unknown-key parity proves neither parser equivalence nor portability. Python has exactly seven deterministic checks: `action_policy`, `precondition`, `object_reachability`, `npc_knowledge`, `forbidden_disclosure`, `quest_stage`, and `quest_stage_monotonicity`. Godot's six state-relative families and seven DISC rules are distinct, non-isomorphic labels; its `QUEST STAGE` display spans two Python quest checks.

The ontology role map binds source declarations to frozen expectations. Table III's `reveal_hint(actor_id,fact_id)` separates view, repair, and authority: the view exposes `tide_marks_hint`/stage-2 policy; $\rho$ sees only $a_t, E_t$, may insert a named predicate, and cannot read `WorldState`.

TABLE III
EIGHT ONTOLOGY ROLES: READ-ONLY SOURCE BINDINGS, NOT LIVE INFERENCE OR AUTHORITY

| Role | Source declaration or relationship example |
|---|---|
| World | Initial object locations place the signal lens in the lamp store. |
| Actor | Captain Mira is the sole eligible actor. |
| Knowledge | Actor knowledge refers to canonical Fact IDs, not permission. |
| Fact | Initial facts include `dock_saved` and `lighthouse_dark`; `keeper_betrayal` is an ID, not payload. |
| Policy | Disclosure policy forbids `keeper_betrayal`; `tide_marks_hint` needs minimum quest stage 2. |
| Quest | The initial quest is at stage 0 with flag `dock_saved`; disclosure preserves stage. |
| Action | `reveal_hint(actor_id, fact_id)` adds Fact/disclosed ID; only tide hint adds relationship memory. Fresh revision $+1$; redisplay unchanged. |
| Trace | Prior and next state hashes, rule and template identifiers, and case ID bind packet/version to evidence; absent fields cannot establish PASS. |

1: $a \leftarrow \text{Propose}(c)$; $j \leftarrow 0$
2: **repeat**
3: $(\text{ok}, E) \leftarrow V(c, a)$; $\text{Record}(a, E)$
4: **if** ok **then**
5: **assert** $V(c, a).\text{ok}$; **return** $\text{Commit}(c, a)$
6: **if** $j = K$ **then return** $\text{Fallback}(c)$
7: $a \leftarrow \text{Repair}(c, a, E)$; $j \leftarrow j + 1$
8: **until terminal**

Fig. 3. Validate–repair–commit. At most $K + 1$ candidate attempts are recorded; success defensively validates before mutation.

Godot commits only all-seven-pass candidates; fresh disclosure increments revision $+1$; redisplay does not.

### B. Fixed-goal NPC integration

Under `npc-goal-recovery/1.1.0`, Codex proposes intent, Python checks public claims/plans, and only Godot writes state. The fixed harbor-restoration/authorized-tide-lead goal is not model-editable. Bindings cover session, epoch, request, scenario, NPC, goal, version, policy/config hashes, revision/state hash, deadline/budget, and issued action. Service: `127.0.0.1:8768`; browser origins: `localhost`, `127.0.0.1:8877`.

*1) NPC-DISC-1 disclosure contract and authority boundary:* NPC-DISC-1's 60 declaration-derived `reveal_hint` cases do not rerun/enlarge historical `verified-run-006`. Generated results/Tables X–XII define strata, bindings, and input-to-check mappings; Godot rechecks service bindings before mutation.

### C. Validate–Repair–Commit

In Fig. 3's dependency-free pseudocode, the repair budget is $K \in \mathbb{Z}_{\geq 0}$ and Record appends in memory; persistence follows terminal consistency. $(\text{ok}, E)$ abbreviates $V$ with $\text{ok} := \bigwedge_{i=1}^{7}(v_i = 1)$; $V(c, a).\text{ok}$ revalidates.

TABLE IV
PER-ERROR-CLASS EDITS OF THE GUIDED OPERATOR $\rho$

| Validator error class | Edit $\delta(e)$ on the candidate |
|---|---|
| Policy precondition omission | insert the named predicate |
| Policy effect omission | insert the named effect |
| Policy effect violation | drop the named effect |
| Unsatisfied precondition | drop the named predicate |
| Unauthorized stage mutation | clear `quest_stage_effect` |
| Unknown action type | no-op (no registered alternative) |
| Unreachable object; knowledge, disclosure, and stage classes | no-op (candidate edit would launder or needs state) |

Repair cannot mutate state; exhaustion preserves it. Typed `counterexample_guided_repair` reads only candidate $a$/errors $E$ (its sole state-derived input), never `WorldState`:

$$\rho(a, E) = a \oplus \bigcup_{e \in E} \delta(e), \tag{3}$$

Table IV's $\oplus$ applies deterministic $\delta(e)$: one field/error entity, at most $|E|$/attempt; conflicting requirements/rejections leave fields unchanged. Fixed points require identical $E$, not convergence. The nondeployable state-reading oracle ignores errors, reconstructs preconditions/effects and both quest-stage fields from state, and never launders violations by deleting required objects/used facts/disclosed facts.

### D. Integrity, Semantic Replay, and Accounting

Unkeyed SHA-256 detects accidental/tested changes, not identity or joint rewriting. Replay checks prior-state/candidate/validation, invalid nonfinal attempts, recomputed final state, and JSONL continuity, never repair. Request→decision→result and first-action/one-receipt bindings do not prevent duplicate IDs or impersonation. Terminal classes are commit, symbolic fallback, adapter failure, and controller failure (state/assignment preserved; no completed proposal trace). Pre-execution run/arm/scenario/seed/model/revision keys plus configuration/input/prior-state hashes bind assignments; aggregation rejects duplicate/missing/unexpected keys and separates failures.

## V. EVIDENCE LANES AND BOUNDARIES

### A. *E1* Offline Conformance Design

PQ1–PQ5 cover immutability, repair, checksum/semantic/linkage/continuity faults, rollback, fail-closed assignment/adapter/manifest accounting, and unknown-key rejection. Results enumerate 13 gate fixtures (1 valid, 12 negative) and 12 repair fixtures; latency/token constants test accounting, not population/detector performance. Packets bind fixtures, assignments, schemas, results, and code.

### B. Archived live-screening materials

Archived live-screening is immutable provenance/hash evidence, excluded from active tables/counts/claims and never transferred to `gpt-6.1-sol`.

TABLE V
EVIDENCE LANES AT A GLANCE: DESIGN, UNIT, COMPARISON, AND CEILING (GENERATED FROM THE EXPERIMENT MATRIX)

| Lane | Design | Unit / budget | Comparison | Ceiling |
|---|---|---|---|---|
| E1 | Offline conformance: authored frozen fixtures, one world | 13 gate; 12 repair/arm; 10 faults; 7 adapter; 3 guards | reject vs blind vs $\rho$ vs oracle | mechanism conformance |
| E3 | KG/ontology simulation: closed-world typed-link holdout | 43 nodes; 106 ref. edges; 24 typed; 210 scores | degree baseline vs 6 fixed strategies | construction result only |
| ENG1 | Engineering-only Godot/Web: engine-local policy mirror, no participants | 4 fixtures; 52 checks; 8 smoke; 5 archetypes | authored fixture paths | engineering-only conformance |

Active lanes are non-interchangeable with different denominators; the historical E2 receipts are excluded from active generation and ENG1 is engineering-only. No active lane establishes population efficacy, model ranking, player experience, retrieval benefit, or commercial-engine performance.

TABLE VI
SIMULATION-ONLY PROPOSAL-STRATEGY EVALUATION ON A FROZEN TYPED-LINK HOLDOUT

| Strategy | Precision | Recall | MRR | Brier | Sem@3 |
|---|---|---|---|---|---|
| S0 degree baseline | 0.000 | 0.000 | 0.282 | 0.315 | 0.556 |
| S2 typed-lexical | 1.000 | 1.000 | 0.944 | 0.131 | 1.000 |

### C. NPC-ENG1/NPC-SCR1: 88 Assignments, 28 Definitions

Stable v5 has 16 engineering assignments and 12 screening recipes ($12 \times 2 \times 3 = 72$; combined $R/P/O = 40/24/24$), covering bootstrap, identity/hash, intent/plan/role/result, negative/stale/duplicate/deadline, and replay. P uses authenticated Codex OAuth without fallback; O is 100% auxiliary-source, not independent gold. The verified-run-006 records retain status, counts, latency, and reproduction information; verified-run-002/003/004 are historical. ENG-X01/SCR-X01, D02/QA-001, C-RESULT, Frozen8/Frozen16, v1–v4, and `native012` are provenance, not C6 evidence.

### D. *E3* KG/Ontology Link-Proposal Simulation

E3 is an engineering-only, non-authoritative closed-world simulation: six authored graph-edge-withheld targets select/summarize among seven fixed strategies, not an untouched or external test. **Scope.** This table reports authored closed-world link recovery, not runtime KG retrieval, semantic completeness, long-horizon contradiction reduction, or evidence for any efficacy claim. The deterministic methods graph contains 43 graph nodes, 106 reference edges, and 24 curated typed edges; all ontology checks pass. The generated table is construction evidence, not OWL/SHACL conformance, runtime retrieval, semantic completeness, or efficacy.

## VI. PILOT RESULTS

### A. Declared-Field Gate Conformance

The loader admits only one valid control plus one isolated instance of each implemented validator code; therefore 13/13 agreement and all 12 implemented codes are *harness construction invariants*, not measured success. Beyond this setup, 12 negative fixtures matched authored expected codes rather than another code, and every noncommit path preserved complete canonical state. This is single-world-state structured-field-oracle conformance, not independent semantic-label accuracy.

TABLE VII
E1 OFFLINE CONFORMANCE: EXACT REPAIR-ARM COUNTS BY REPAIRABILITY CLASS (DESIGNED FIXTURES)

| Arm | Info source | Commits / cases | | | Mean edits[e] |
|---|---|---|---|---|---|
| | | G-rep.[f] | O-only[f] | Irrep.[f] | Fields |
| Rejection-only ($K$=0) | none | 0/5 | 0/1 | 0/6 | — |
| Blind unchanged retry | none | 0/5 | 0/1 | 0/6 | — |
| Validator-error-guided $\rho$ | error set $E$ | 5/5 | 0/1 | 0/6 | 1.0 |
| State-reading oracle | state+policy | 5/5 | 1/1 | 0/6 | 1.0 |

[e] Mean changed candidate fields over committed repairs (minimality proxy); failed arm executions left the state unchanged. E1 uses 12 initially invalid designed cases: rejection-only/blind 0/12, 0/12; validator-error-guided $\rho$ 5/12; state-reading oracle 6/12 commits. [f] Commits/cases. G-rep. = guided-repairable, O-only = oracle-only, Irrep. = irreparable (frozen repairability classes); these are not population efficacy or live-model quality.

TABLE VIII
E1 OFFLINE CONFORMANCE: FROZEN-PILOT CHECKS AND TERMINAL OUTCOMES (ROWS ARE NOT COMPARABLE)

| Check or terminal family | Exact outcome | Interpretation |
|---|---|---|
| Gate fixture agreement | 13/13 | authored-oracle agreement |
| Known-boundary acceptance | 2/2 | encoded limit, not a safety pass |
| Closed unknown-key rejection | 1/1 | proposal/replay parser parity |
| Detectable integrity-fault rejection | 10/10 | designated check operations |
| Known provenance-boundary acceptance | 1/1 | repair provenance unauthenticated |
| Adapter terminal classes | 1 commit; 1 fallback; 5 failures / 7 assigned | complete classification of 7 assignments |
| Manifest-guard rejection | 3/3 | injected faults failed closed |

Rows have different denominators and favorable directions; they must not be ranked as rates. Every value is an exact audit fact for designed fixtures.

### B. Repair-Arm Comparison on Designed Fixtures

Frozen classes for 12 initially invalid designed cases are 5 guided-repairable, 1 oracle-only, and 6 irreparable. Rejection-only and blind unchanged-candidate retry each committed 0/12. $\rho$ committed 5/5, 0/1, and 0/6, respectively, editing a mean of 1.0 fields per commit. It reads only the prior candidate and typed validator errors $E$, its sole state-derived input; failed arms preserve canonical state. The state-reading reference callback committed 5/5, 1/1, and 0/6, respectively. Blind never converts errors, $\rho$ handles only payload-sufficient cases, and the oracle adds state knowledge. The 5 committed guided cases initially carried `POLICY_PRECONDITION_OMISSION`+`UNSATISFIED_PRECONDITION`, `POLICY_EFFECT_OMISSION`, `POLICY_EFFECT_VIOLATION`,

UNSATISFIED_PRECONDITION, POLICY_QUEST_STAGE_EFFECT_VIOLATION, respectively, and each became valid through the corresponding Table IV edit $\delta(e)$ alone. Frozen-fixture operator conformance, not live-model repair quality.

### C. Integrity, Boundaries, and Assignment Accounting

All 10 prespecified detectable faults were rejected by designated checks (10/10); without stable typed detector codes, no detector-layer attribution follows. A known provenance-boundary fixture changed the invalid precursor before the same recorded valid repair, recomputed the checksum, and passed replay (1/1 undetected): replay neither authenticates repair generation nor blocks writers recomputing unkeyed checksums. Two open sentinels—unextracted narrative disclosure and simultaneous candidate/policy omission of a required object—were accepted at the encoded layer (2/2), with 0 safety passes. The former unknown-top-level acceptance became a closed regression: proposal and replay parsers rejected a complete 12-field candidate with one unknown key (1/1), showing parser rejection parity, not semantic safety. Seven adapter/accounting assignments yielded 1 commit, 1 symbolic fallback, and 5 adapter failures; synthetic telemetry was populated and propagated for 2/7 (schema-pinned accounting, not measurement). All 3/3 injected duplicate/missing-assignment guards failed closed; these are raw counts from synthetic offline fixtures.

### D. Fixed-Goal NPC Operation (NPC-ENG1/NPC-SCR1)

Mira delivers the authorized tide lead after harbor-signal restoration. Python computes bounded BFS plans from typed intent/public observations; players acquire/install, while the NPC commits only the authorized `reveal_hint`. Godot synchronously checks session/epoch/request/goal/policy/revision/hash before calling the existing writer without an await; results/new observations choose the next plan, not single-candidate validator-error repair.

NPC-ENG1 has 16 deterministic R-arm engineering cases from actual engine snapshots (assigned_failure=2, completed=14); NPC-SCR1 retains all 72 P/R/O assignments and failures, 24 per arm (assigned_failure=12, completed=60). P records 24 authenticated Codex CLI exec attempts, 24 returned decisions, and 4 assigned failures—not remote billing-call counts. Requested model: unknown (null); recorded decision-model identifier: unknown (null). The latter is the configured selector, not a provider revision; provider revision is unavailable. Null identifiers remain unknown; global defaults are not evidence. R uses authored rules. O is a protocol-declared auxiliary under `same_model_family_auxiliary`. Actual model-family matching is unverified. No source is independent; O is neither an oracle nor semantic gold. Table IX gives bounded protocol-conformance fixed-case counts, not recovery-efficacy, superiority, or generalization estimates; efficacy pending, tokens/cost unmeasured. A read-only CLI and isolated working directory were used; tool-free inference and adversarial confidentiality were not established. Immutable verified-run-006 is revalidated against archived source bytes dated 2026-10-17 (revision fa9fb8a); this historical packet result does not certify the current checkout NPC runtime or current release authority. Verified pre-authored stale-control ACK rejections are P=4, R=4, O=4 per arm (SCR-T01 (stale_revision), SCR-T02 (stale_policy)); 12 instances had all corresponding negative-control gates passed. Assigned failures remain in the denominator, not method-performance outcomes; other assigned failures (0) remain unclassified. Replay validation is limited to the exported epoch (`intent_lineage_scope=epoch_local_only`); it does not prove intent continuity across epochs (`cross_epoch_intent_continuity_proven=false`). The opaque-anchor exception allows only an adjacent leading observation request/decision pair in a positive-epoch snapshot; epoch-zero orphans, other observation requests before intent commitment, and later intent mismatches remain rejected.

TABLE IX
NPC-SCR1 FIXED-CASE EXECUTION COUNTS, SEPARATE FROM THE 16 NPC-ENG1 CASES; ALL ASSIGNED FAILURES REMAIN IN THE DENOMINATOR.

| Arm | Assigned | Complete/fail | Goal reached | Recovery E/U |
|---|---|---|---|---|
| P | 24 | 20/4 | 4 | 24/0 |
| R | 24 | 20/4 | 4 | 24/0 |
| O | 24 | 20/4 | 4 | 24/0 |

E: recovery eligible; U: unreachable; NA: unmeasured. Observed goal counts are not recovery rates. Valid replay does not turn a failed assignment into a completed one. O has zero independent sources.

### E. Frozen `reveal_hint` contract and disclosure evidence

Only the frozen expectation packet `sl-npc-disclosure-expectations-v1` (v1.0.0) and four result files are consumed. Packet SHA-256 `9920aa834910a3ee35ed4446de4fdba42bb67dda71b07a93bccfc7a0b2e09028` was checked against its sidecar. The declared normal stratum has 48 cases under `1 eligible actor x 4 declared fact IDs x 3 declared stage values x 2 knowledge values x 2 prior-disclosed values`; boundary 4 and malformed 8 are excluded. Trace expectations match packet case IDs, not gate output.

Content-first explanation maps typed input (operation, actor, fact, stage, knowledge, prior disclosure) through ontology entities/relations and authored rules to checks, an allowed/denied decision, and canonical state effect. A permitted first display adds only the requested fact (plus the declared relationship-memory entry for the tide hint), preserves stage, and increments revision once; denial/redisplay preserves canonical state. Explanatory only: no hidden facts, natural-language implications, or live capture.

Runtime declaration: `sl-npc-disclosure-expectations-v1:runtime-disclosure` / v1.0.0, provenance `frozen_expectation_packet`. Scope contains only four canonical fact IDs (dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint) and eligible actor `captain_mira`. Rule IDs: DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007;

TABLE X
FROZEN REVEAL_HINT CONTRACT STRUCTURE, CONTENT, RULE IDS, AND RESULT PROVENANCE.

| Field | Value |
|---|---|
| Operation | reveal_hint |
| Scenario | sealed-lighthouse-v1 |
| Normal denominator | 48 |
| Boundary | 4 |
| Malformed | 8 |
| Canonical facts | dock_saved, keeper_betrayal, lighthouse_dark, tide_marks_hint |
| Eligible actor | captain_mira |
| Ontology | sl-npc-disclosure-expectations-v1:runtime-disclosure |
| Rule IDs | DISC-001, DISC-002, DISC-003, DISC-004, DISC-005, DISC-006, DISC-007 |
| Templates | mira-safe-deflection-v1; sl-rc-003-authorized-hint |

Expectations are not derived from gate output; denominators are read from the frozen declaration.

approved templates only `mira-safe-deflection-v1` and `sl-rc-003-authorized-hint`.

Python evaluator materializes the same cases/expectation object in three isolated arms, recording semantic outcomes, check traces, errors, before/after state SHA-256, rendered template IDs, and replay equality. Godot separately owns the existing `reveal_hint` writer and canonical mutation at the game-execution boundary; this producer neither reconstructs nor replaces Godot state. Traceability binds operation `reveal_hint`, scenario `sealed-lighthouse-v1`, actor `captain_mira`, and bound source hashes. Execution-observed hashes match the current source set. 3 source hash(es) differ from the frozen baseline (game controller, lighthouse state machine, ontology-signals interview source as a non-identical public derivative). The packet hash is preserved; source-specific hashes and reproduction information remain in the authors' producer record, which is not deposited. The public derivative is not byte-equivalent to the private historical source; it is publication-materialization provenance only and implies no new gameplay or model observation.

Table XI reports the three arms over the same 60 cases with separate normal/boundary/malformed denominators; unchecked violations are intentional isolated observations. The checked general-rule and ontology-gate arms have identical outcomes on all 60 of 60 authored cases. The rule-change fixture set contains no discriminating case between these arms: all 2 fixtures and their 4 paired cases match. This is an authorization-conformance limit under the shared frozen rules, not general ontology equivalence or ontology superiority. Figure 4 distinguishes E/R (expected-match/replay cases), C (check-record PASS/FAIL), and NR (cases with no reported checks); NR is not skipped, pass, or out-of-scope. Table XII maps rule-change fixtures and paired cases to disposable policy clones; the shipped scenario is unchanged.

Claims are limited to bounded engineering conformance: no human gold labels, access-separated holdout, efficacy, superiority, generalization, player benefit, model latent state, or cost. The authors' producer record binds input identities and hashes; it is held locally and no reproduction commands are provided here.

**ENG1.** Legacy Godot 4.7.1 replays authored rejection/commit against prior/offline hashes; presentation-only captures show board/hints/coach/chain prompt and dashboard checks. Core 52/52 plus four extensions establish engine-local conformance, not transport, participants, narrative, usability, or efficacy.

### *F. Single-task Codex development observation*

Frozen `ontology-map-20261023-01` covers one read-only bilingual dashboard task ($n$=1, independent repeats

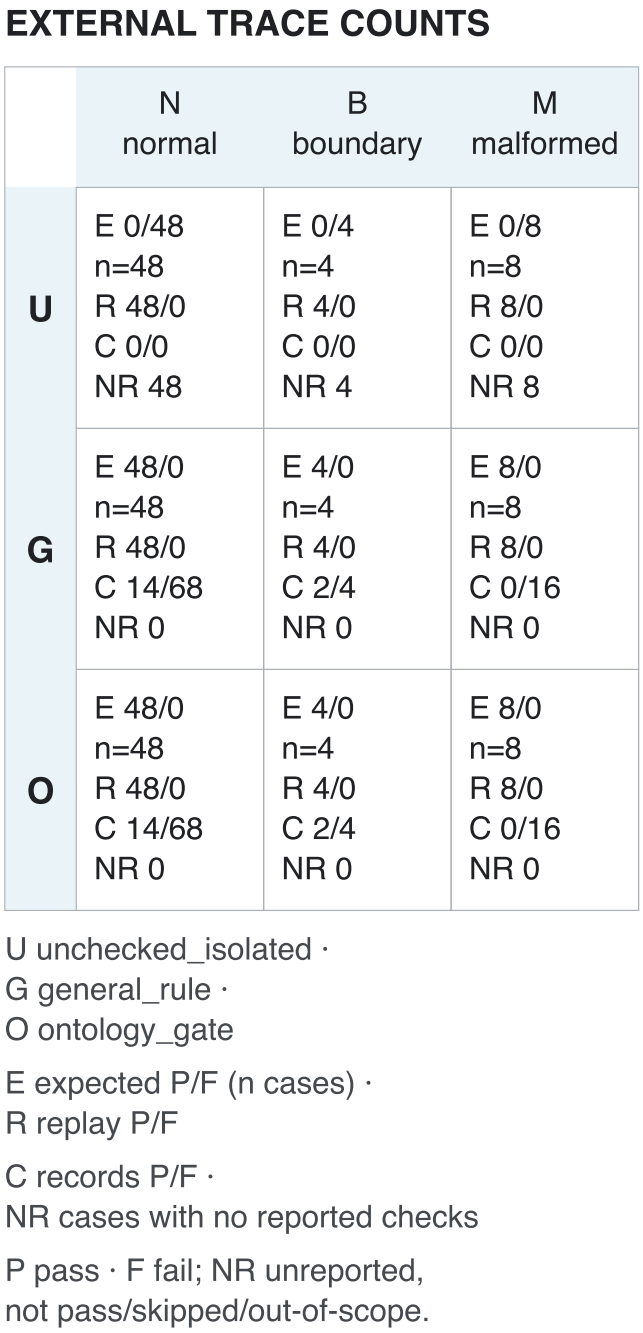



Fig. 4. NPC-DISC-1 counts by arm/stratum. E/R: expected-match/replay; C: policy-check PASS/FAIL; NR: no reported check, not skipped. Packet-bound engineering counts measure neither latent state, free-language leakage, ontology superiority, nor player benefit.

TABLE XI
RAW-TRACE OBSERVATIONS FOR THE THREE DISCLOSURE ARMS WITH SEPARATE NORMAL, BOUNDARY, AND MALFORMED DENOMINATORS.

| Arm | $n$ (N/B/M) | Allow/Deny | Semantic mismatch | State/Display fail | Replay fail | Unique unsafe |
|---|---|---|---|---|---|---|
| unchecked_isolated | 48/4/8 | 60/0 | 60 | 0/0 | 0 | 51 |
| general_rule | 48/4/8 | 16/44 | 0 | 0/0 | 0 | 0 |
| ontology_gate | 48/4/8 | 16/44 | 0 | 0/0 | 0 | 0 |

N/B/M = normal/boundary/malformed. Semantic mismatch compares the complete expected outcome; unique unsafe cases are the union of unauthorized allows and redisplay mutations. Permitted fresh-disclosure representation mismatches alone are not unsafe. Unchecked observations do not establish efficacy, superiority, or generalization.

TABLE XII
RULE-CHANGE FIXTURE TRACEABILITY AND EDIT SCOPE FOR PAIRED CASES; THE SHIPPED SCENARIO IS NOT MODIFIED.

| Fixture | Mutation | Paired cases | Changed paths | Missed cases | Match |
|---|---|---|---|---|---|
| RC-001-stage-gate-2-to-1 | disclosure_policy.stage_gates[0].minimum_stage 2→1 | N-043, N-044 | disclosure_policy.stage_gates[0].minimum_stage | none | PASS |
| RC-002-forbid-tide-marks-hint | disclosure_policy.permanently_forbidden ['keeper_betrayal']→['keeper_betrayal', 'tide_marks_hint'] | N-047, N-048 | disclosure_policy.permanently_forbidden[1] | none | PASS |

Each fixture applies the same mutation to general_rule and ontology_gate. PASS requires no missed case and mutation_scope_preserved=true.

0), not an NPC trial/cohort. Six native CLI invocations (initial + five resumes; one read-only/no-edit) form one case. `turn_context.model=gpt-6.1-sol` is runtime selection, not provider attestation; provider identity/revision/seed/cost are `null`, and model-only latency is unavailable. Wall-clock 5942.665 s and CLI 263.32 s are neither inference latency nor the future endpoint.

Invocation six fixed inherited 660 px mobile-table minimum width: geometry RED (2 failed/35 passed), followed by 62/62 map/signal/Web checks (including cell/paragraph geometry) and a successful Godot Web build. Browser attempt 10 passed; 01–08 failed setup/routes; 09 passed older gates with clipping. Final $(\text{seq}, t_{\text{ms}})$: stage-0 denial $(68, 19468)$ (equal hashes), fresh disclosure $(239, 58234)$ (different), authorized redisplay $(249, 59793)$ (equal). At 1440×900/390×900, Enter, inspection/frozen-comparison invariance, and zero pre-navigation console/page/network errors/warnings were observed. A later rerun of the same driver on the production dashboard is recorded separately as overall **NOT_PASS**: the headed run stopped at the unchanged yaw guard, and the headless run passed seven component gates but logged four pointer-lock page errors. It does not overturn the dated attempt-10 PASS, which was judged under the acceptance set then in force. This bounded integration is neither full-game creation, Python–Godot authorization round trip, nor a model outcome. Privacy/source-binding failures and same-thread repairs add no $n$; Jev is `proposal_only`, without label/approval authority.

### G. Fixed-game native runtime comparison

This public packet retains 8 assigned rows for one authored quest (`sealed-lighthouse-v1`), with seed 20261024 and two repeats per selector/representation cell. System-endpoint successes are 5/8; observed native executions are 7. Host invocations are summed from actual receipts (23), and observed repairs sum to 0. Committed transitions total 16, of which unsafe are 0; semantic replay valid/invalid/NR is 7/0/1. Fresh-engine reexecution disposition counts are VALID=6; INVALID=0; NOT_RUN=1; NOT_OBSERVED=1. Observed elapsed time is n=7; 48.19/41.09 seconds (mean/median), and preparation time is n=7; 6.67/6.30 seconds. Unreported values are NR, not zero.

The first row is retained as a harness JSON-serialization technical failure and is not attributed to GPT-6.1 capability. At least 2 GPT-6.1 ontology row(s) retain a response-timeout failure under the frozen 60-second bound. Models provide typed-intent proposals; the deterministic Python planner/checker and Godot commit retain authority. The table is therefore a technical observation for one quest, not an efficacy, superiority, noninferiority, general ontology-consistency, or player-benefit claim. Semantic record-link replay is not fresh-engine reexecution. Provider identity/revision/cost and provider-request counts remain NR outside the public observation boundary.

### H. Separate exploratory signal example

An exploratory receipt records one synthetic `missing_effect` case, separate from the preceding results and the prospective pilot. The shared proposal omitted the required `signal_restored` effect. The deterministic control repaired it in one attempt, committed, and reached the independently specified objective that the committed state contain that fact.

The two model conditions each had one recorded attempt and received signals of different information content: the detailed signal identified the error, missing effect, and repair hint, whereas the generic signal only said `Action rejected.` The detailed response added the effect and, upon revalidation, committed and reached the objective; the generic response left the effects empty, failed revalidation, and fell back without reaching the objective. This contrast is descriptive for one case, not an information-matched comparison, causal or statistical evidence, an overall study result, or an answer to questions A or C. The requested model was `gpt-6-luna`; the identity of the responding model, cost, provider request count and retries are unknown. Public availability of the inputs has not been verified. The receipt does not contain the complete prompts

TABLE XIII
FIXED-GAME NATIVE RUNTIME COMPARISON (N=2 PER CELL). ALL EIGHT ASSIGNMENTS, INCLUDING THE HARNESS FAILURE AND TIMEOUTS, REMAIN IN DENOMINATORS.

| Requested selector | Form | Goal/assigned | Observed exec. | Host calls | Unsafe/committed | Elapsed median (s; n) | Fresh engine |
|---|---|---|---|---|---|---|---|
| gpt-6.1-sol | general_rule | 1/2 | 1 | 4 | 0/3 | 45.91 (1) | 1/0/0/1 |
| gpt-6.1-sol | ontology | 0/2 | 2 | 3 | 0/1 | 72.35 (2) | 1/0/1/0 |
| gpt-5.6-sol | general_rule | 2/2 | 2 | 8 | 0/6 | 40.24 (2) | 2/0/0/0 |
| gpt-5.6-sol | ontology | 2/2 | 2 | 8 | 0/6 | 33.11 (2) | 2/0/0/0 |

Fresh engine counts follow VALID/INVALID/NOT_RUN/NOT_OBSERVED order. NR is unreported, not zero. Overall counts and the separate semantic replay are reported in the text.

or the executed code version, so it does not support exact reproduction of these runs.

### *I. Separate synthetic development pilot*

For each requested selector, `gpt-5.6-luna` and `gpt-5.6-terra`, the separate synthetic development pilot (the v2 evidence cohort) planned 10 assignments: the first detailed `missing_precondition` assignment was recorded as `FAILED` with `protocol_error`, the other 9 remained pending, and there were 0 valid model responses. Collection stopped after that first failure. Raw error text was not retained because it could contain sensitive data; the cause is unknown. Whether a request was transmitted or a provider response received, the actual provider model identity, and cost are unverified. These fictional development cases are neither a holdout nor a human evaluation; they are separate from the preceding single-case observation and other studies. They do not establish a completed performance comparison, a confirmatory result, a model failure or lack of support, or a statistical effect. Separate selector outcome records preserve these assignment statuses.

A separate diagnostic cohort (the v2 diagnostic cohort) planned 10 assignments per selector. Terra recorded the first detailed `missing_precondition` assignment as `FAILED` with `protocol_error` and left 9 pending. Luna's first detailed assignment remains `IN_PROGRESS` in the case record, with 9 other rows pending; its aggregate status counts all 10 as pending. The operator reported that the process was absent; transmission and the reason for termination are unknown, and the unfinished row is not a success. The diagnostic outcome records are not pooled with the earlier attempts and do not establish model performance or lack of support.

**Historical bookkeeping.** Completed/failed/pending assigned-row counts are: original pilot $0/7/3$ (the seven failures are technical or interrupted attempts, not seven demonstrated unsuccessful model answers); four initial selector/native files $0/0/40$; initial v2 pair $0/0/20$; startup diagnostics $0/2/18$; v3 pair $0/0/20$; v4 Luna $10/0/0$ (detailed goal $4/5$, generic goal $4/5$), and Terra $0/1/9$. The v2 evidence counts (one failed and nine pending rows per selector) and the v2 diagnostic counts above remain unchanged, including Luna's `IN_PROGRESS` snapshot. These are file-bookkeeping groups, not pooled independent samples or provider requests; pending rows establish neither requests nor model failures, and v4 and v5 remain separate. Actual served identities, costs, and provider request counts remain unknown. The safe historical projection retains per-cohort counts and local-original digests, not raw source access or standalone reproducibility.

### *J. Fresh v5 paired-selector development collection*

This fresh v5 collection is separate from all earlier incomplete and diagnostic cohorts. Each requested selector received five synthetic development fixtures under detailed and generic feedback (10 assignments each), in nominal fixed serial order Luna then Terra. All 20 assignments have a received response and none remain pending; response receipt is not task success. Table XIV separates strict validator-gate acceptance, independently specified objective success, and schema failure, each over all five assigned cases per cell. The collector applied the strict parser and deterministic checks, then evaluated the objective separately. Of the 20 assignments, 8 passed the gate, 7 reached the objective, and 10 failed schema parsing; these descriptive counts do not identify a causal effect.

Under the frozen collection amendment, a schema-invalid response remains a failed assignment and a goal failure, but collection continues after proven adapter completion; genuine transport or safety failures still stop collection. Gate acceptance is not objective success: one accepted Luna detailed case did not reach its goal. The five fixtures are `missing_precondition`, `missing_effect`, `unauthorized_objective`, `forbidden_disclosure`, and `unsatisfied_dependency`. In `unauthorized_objective`, the frozen `signal` policy permits no effects although the common goal requires `signal_restored`; Luna's accepted detailed response removed the forbidden effect and therefore did not reach that goal. All five assignments, including this policy/goal-conflict case, remain in each denominator. The small synthetic development set is neither held out nor a human or causal study, and detailed versus generic feedback has unequal information content. Actual served model identity/revision, provider request count, and cost remain unknown. These observations establish neither selector superiority, ontology efficacy, nor confirmation of A or C.

The public-safe outcome projections, held by the authors and not deposited, retain all case outcomes, prompt hashes and executable hash/version, while omitting the local executable path and disclosing the original-file digest and omitted field. Frozen raw outcomes and approvals remain local and unchanged. A common immutable source snapshot preserves seven executed source files; it is provenance evidence, not a standalone

TABLE XIV
SEPARATE V5 SYNTHETIC DEVELOPMENT COLLECTION. EVERY DENOMINATOR INCLUDES ALL FIVE ASSIGNED CASES, INCLUDING SCHEMA FAILURES; RECEIVED RESPONSES DO NOT IMPLY SUCCESSFUL TASKS.

| Requested selector | Signal | Received | Strict gate accepted | Goal | Schema failure |
|---|---|---|---|---|---|
| `gpt-5.6-luna` | Detailed | 5/5 | 4/5 | 3/5 | 1/5 |
| `gpt-5.6-luna` | Generic | 5/5 | 3/5 | 3/5 | 0/5 |
| `gpt-5.6-terra` | Detailed | 5/5 | 0/5 | 0/5 | 5/5 |
| `gpt-5.6-terra` | Generic | 5/5 | 1/5 | 1/5 | 4/5 |

runnable distribution. For this v5 collection, successful parsed response text is retained, but raw CLI stdout and stderr were not retained because they could contain credentials or sensitive server data; byte counts, digests, and event summaries remain. The ten failed assignments record `CandidateParseError` as the evaluation error with null evaluations, so the retained artifacts do not establish their content-level causes. They remain in all assigned denominators; neither receipt nor failure establishes provider identity or inherent model inability.

Figure 5 exposes the gate result as an explicit signal, not an affect estimate or permission: the detailed arm carries the failed check and repair hint, while the generic arm carries only rejection. Both responses face the same strict parser, identity check, and seven predicates. This v5 collector uses zero further repair calls, unlike the general bounded repair loop. Figure 6 follows every assignment through receipt, parsing, gate acceptance and the independent goal; it also locates the schema failures and the accepted commit that missed the goal.

### *K. Prospective two-track pilot*

The requested panel is Codex `gpt-5.6-sol`, Codex `gpt-6.1-sol`, and Antigravity `gemini-3.8-flash-medium`; these are requested selectors, not attested served identities. Medium effort is nominal across providers, not equal compute. Runtime crosses three targets with two identical-information representations and ten fresh resets (60 units, 180 s per episode). Bounded development crosses three targets, minimal input versus public-repository exploration, the same two representations, five tasks, and two fresh runs (120 units, 900 s per assignment). Budget-bounded independently validated success is primary; hard safety remains a separate gate. The protocol preserves seeds, order, reset identifiers, repair and dispatch limits before collection.

**Prospective pilot status: not a model comparison.** The frozen study `prospective-pilot-20261103` plans 180 units (60 runtime and 120 development) over three requested targets. 180 are blocked, 0 assigned, 0 attempted and 0 executed, with 0 native process starts. Status `NOT_RUN/BLOCKED`; scientific conclusion `NOT_MET`.

Success rate: null. Observed cost: null (known subtotal null; 0 admitted units with unknown cost). Effect estimate: null. A null is an absent measurement, never zero.

Blocked prerequisites, counted per unit: `BLOCKED_ACTIVATION_ABSENT` (180), `BLOCKED_IDENTITY_FORMAT_UNOBSERVED` (180), `BLOCKED_ISOLATION_UNPROVEN` (180), `BLOCKED_LIVE_PATH_NOT_IMPLEMENTED` (180), `BLOCKED_NO_CREDIT_ENFORCEMENT` (180), `BLOCKED_PUBLIC_DERIVED_UNAPPROVED` (180), `BLOCKED_SCHEMA_ACCEPTANCE_UNOBSERVED` (180), `BLOCKED_TARGET_UNVERIFIED` (60), `BLOCKED_TASK_FREEZE_INCOMPLETE` (180).

The earlier eight-row 5/8 pilot, the stopped identity extension, the Ollama lane, the deterministic fixtures and the n=1 results are separate lanes. They are neither pooled nor re-reported here.

After the blocked collection snapshot, independent offline checks accepted two correct variants and rejected the baseline and plausible-wrong variant for each of five tasks; the task/corpus leakage check passed over 9303 files. These are fixture controls, not model runs, and do not retroactively admit blocked units. Included public tests and cross-language semantic mirrors are disclosed; the endpoint is bounded repair, not whole-game creation or retrieval of intact target originals. The later task freeze does not establish response-bound identity, no-credit enforcement, isolation, or new activation. The hypothesis that newer agents explore every relevant skill, source, and reference remains untested; permitted exploration is not observed exhaustive exploration. Current-source screenshots establish presentation/state correspondence only.

## VII. DISCUSSION, THREATS, ETHICS, AND ARTIFACT SCOPE

Our guided semantic repair operator $\rho$ [48], [55], [56], [57] uses deterministic pre-execution errors; the oracle-only arm reads state. Rule agreement/edit-scope diagnostics are neither capability nor ontology superiority. Implementation confound: row 1 failed under controller-v3 before a model response; rows 2–8 used controller-v4 after a one-line JSON-boundary correction with unchanged model-visible inputs, so selector/representation differences are not model-attributable.

The fixed-game pilot is bounded native evidence; no ontology-efficacy test ran. A (model/stack consistency) and C (fresh-reference noninferiority) remain unexecuted and require budget-bounded completion under fixed source/policy, sealed assignments, exact identity/revision fields, and separate safety/replay accounting; it answers neither. Endpoint time is request/evidence receipt to the first receipt passing frozen mandatory checks (time-to-first-all-predeclared-validation-pass); unfinished/timeout/failure/stage-not-reached runs have no successful time. The proposed endpoint uses continuous task-clock seconds.

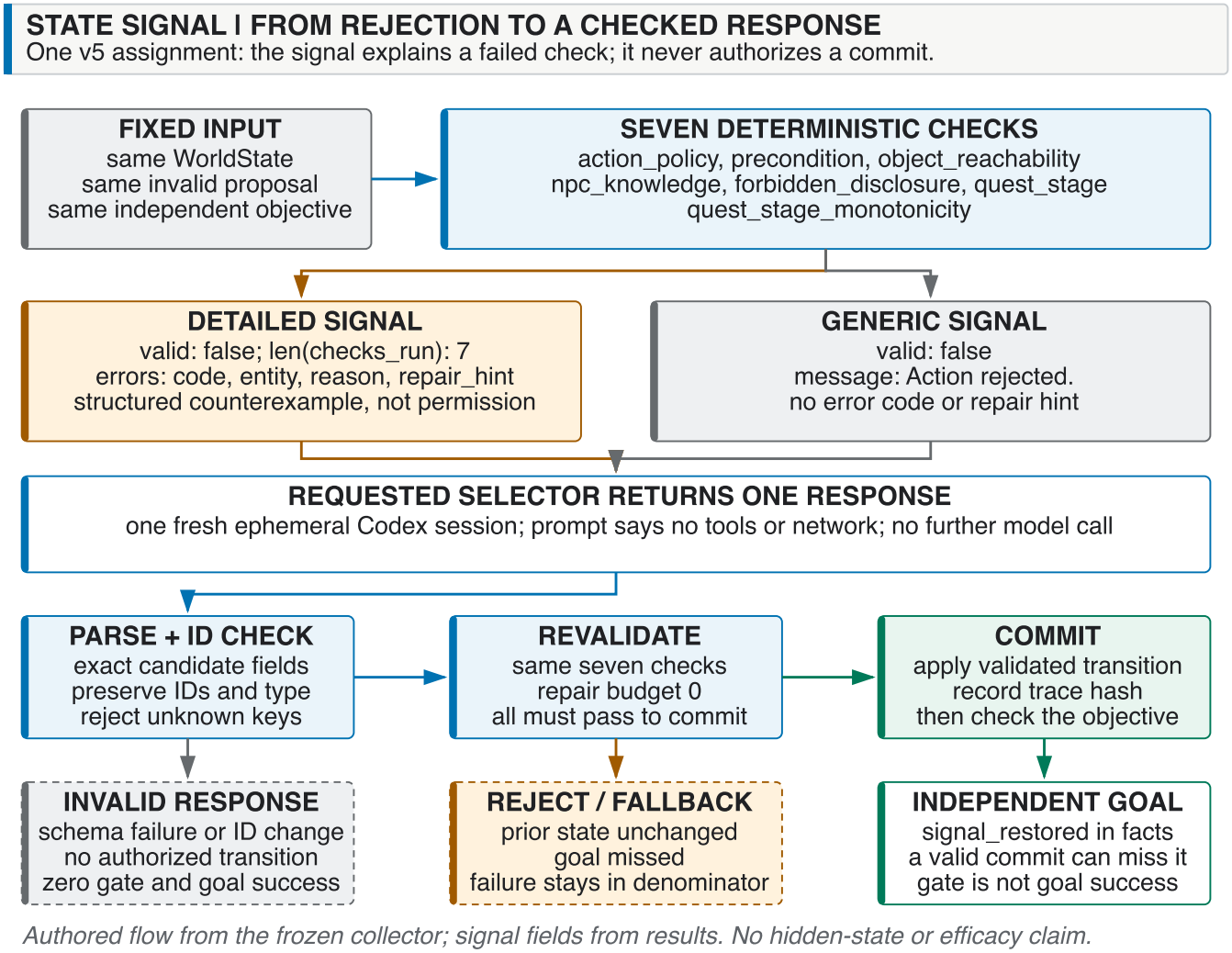



Fig. 5. Source-derived v5 state-signal path. The structured $(\mathrm{ok}, E)$ check result can guide a repair, but only revalidation authorizes mutation. Generic feedback omits error details. The objective is checked separately from gate acceptance; this is an authored mechanism diagram, not efficacy evidence.

The separate identity-required extension in Appendix B stopped at missing served-model evidence; no identity-qualified new comparison was obtained.

**Threats and boundaries.** The native pilot has one authored quest and two repeats per cell; comparator power, neutral complete-attempt inventories, and independent traceability scores are absent. State consistency does not imply behavioral consistency [58]. It also does not establish narrative coherence. Predicate/role/Godot/DISC sets are not latent states; declarations are not human gold/holdouts; lanes retain separate denominators. No participant, affect, comprehension, agency, enjoyment, immersion, benefit, population, safety-rate, model-ranking, or commercial-performance inference follows. Hashes do not authenticate; replay equality is not semantic correctness. Human/affect studies require ethics review, consent, compensation disclosure, minimization, retention/deletion, opt-out, and calibrated judges [40], [41], [42], [43].

## VIII. Conclusion

TRACE-RPG combines bounded C1–C5 authorization/repair/replay/accounting with a source-bound bilingual observer. The Codex case/native pilot retain failures and boundaries, not model-performance evidence. Whole-game, time-to-first-all-predeclared-validation-pass, traceability, confirmatory A/C, and repeated-cross-model evaluation remain future; historical cohorts stay unpooled.

## Disclosure of AI Usage

LLMs assisted prose/code/tests/citations; hosted candidates were labelled. Curated AI art/desk mesh has file provenance; snapshots/hashes/ledger are presentation-only. Authors checked sources and accept responsibility; NPC inputs/tables retain assignment/protocol/schema hashes and unknown token/cost/model-revision fields.

## Data and Code Availability

The preserved materials comprise implementation and build source, frozen packets, the Godot slice, deterministic evalua-

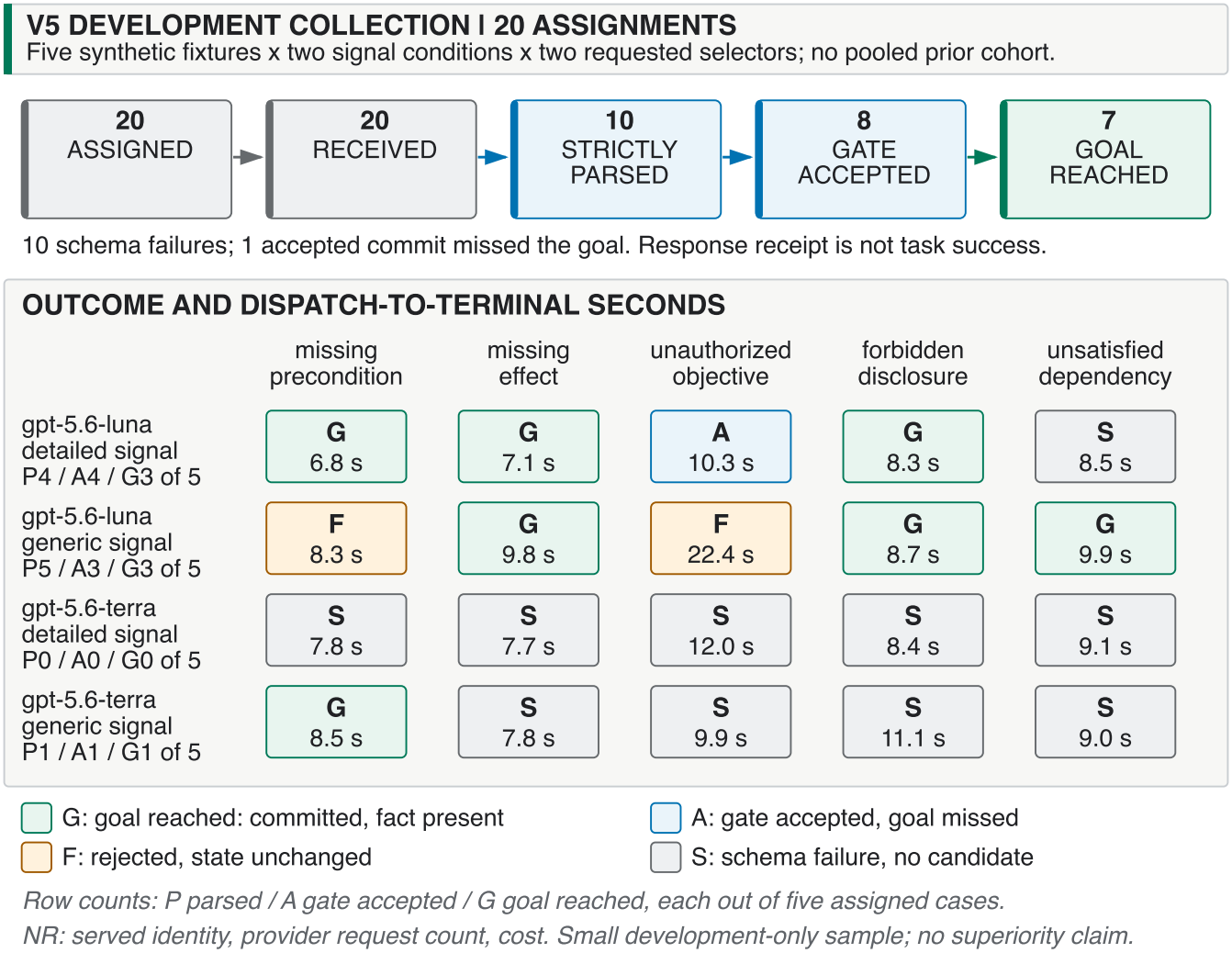



Fig. 6. Frozen v5 collection flow and all 20 per-assignment outcomes, derived from the same receipts as Table XIV. Ten schema failures remain in the denominator; eight accepted candidates yield seven goals. Seconds measure dispatch to terminal, including failed responses, not model-only latency. Requested selectors and unequal-information signals do not establish model superiority or ontology efficacy.

tion, and machine-checked contribution, reference-topic, and experiment matrices. The development-case record documents the bilingual dashboard task. The authors' public-safe projection of the native comparison (held by the authors, not deposited) contains results, public traces, and an integrity audit; the runtime tables and figure derive from those records. A proposed rubric, codebook, source manifest, and future whole-game/repeated-cross-model plan describe unmeasured work. Frozen disclosure inputs and signal/NPC artifacts support the corresponding engineering analyses; earlier records are historical receipts, not current-runtime proof. No reviewer archive or DOI deposit is claimed. This paper provides no repository or deposit identifier for independently locating or accessing the public projections; their external accessibility is therefore not established here. The accompanying traceability supplement presents the proposed, unmeasured rubric in both languages. Separate identity-admission outcomes are not pooled with the original eight rows.

## REFERENCES


[1] M.-A. Côté, Á. Kádár, X. Yuan *et al.*, "TextWorld: A learning environment for text-based games," in *Computer Games*, ser. Communications in Computer and Information Science. Springer International Publishing, 2019, vol. 1017, pp. 41–75.

[2] R. Wang, P. Jansen, M.-A. Côté, and P. Ammanabrolu, "ScienceWorld: Is your agent smarter than a 5th grader?" in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2022, pp. 11 279–11 298.

[3] N. Weir, R. Thomas, R. d'Amore, K. Hill, B. Van Durme, and H. Jhamtani, "Ontologically faithful generation of non-player character dialogues," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024, pp. 9212–9242.

[4] T. Ashby, B. K. Webb, G. Knapp, J. Searle, and N. Fulda, "Personalized quest and dialogue generation in role-playing games: A knowledge graph- and language model-based approach," in *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. ACM, 2023, pp. 1–20.

[5] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. ACM, 2023, pp. 1–22.

[6] Y. Shao, L. Li, J. Dai, and X. Qiu, "Character-LLM: A trainable agent for role-playing," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2023, pp. 13 153–13 187.

[7] N. Akoury, Q. Yang, and M. Iyyer, "A framework for exploring player perceptions of LLM-generated dialogue in commercial video games," in *Findings of the Association for Computational Linguistics: EMNLP 2023*. Association for Computational Linguistics, 2023, pp. 2295–2311.

[8] M. Hochreiter, S. Kriglstein, and G. Wallner, "Beyond pre-defined scripts: Player perceptions on generative non-player character dialogues," in *Proceedings of the 31st International Conference on Intelligent User Interfaces*. ACM, 2026, pp. 2004–2018.

[9] M. Yin, H. Zu, W. Gu *et al.*, "How contextualized generative AI shapes player experience in games," *Entertainment Computing*, vol. 58, p. 101194, 2026.

[10] V. Figueiredo and D. Elumeze, "Symbolically scaffolded play: Designing role-sensitive prompts for generative NPC dialogue," 2025, preprint.

[11] G. N. Yannakakis and J. Togelius, "Experience-driven procedural content generation," *IEEE Transactions on Affective Computing*, vol. 2, no. 3, pp. 147–161, 2011.

[12] D. Melhart, D. Gravina, and G. N. Yannakakis, "Moment-to-moment engagement prediction through the eyes of the observer: PUBG streaming on Twitch," in *FDG '20: Proceedings of the 15th International Conference on the Foundations of Digital Games*. ACM, 2020, pp. 1–10.

[13] Z. Wang, Q. Guo, R. Singh, and X. Hu, "Do vision language models understand human engagement in games?" 2026, preprint.

[14] L. Fan, G. Wang, Y. Jiang *et al.*, "MineDojo: Building open-ended embodied agents with internet-scale knowledge," in *Advances in Neural Information Processing Systems*, vol. 35. Neural Information Processing Systems Foundation, 2022, pp. 18 343–18 362.

[15] D. Pérez-Liébana, J. Liu, A. Khalifa, R. D. Gaina, J. Togelius, and S. M. Lucas, "General video game AI: A multitrack framework for evaluating agents, games, and content generation algorithms," *IEEE Transactions on Games*, vol. 11, no. 3, pp. 195–214, 2019.

[16] X. Liu, H. Yu, H. Zhang *et al.*, "AgentBench: Evaluating LLMs as agents," in *International Conference on Learning Representations*, 2024, pp. 52 989–53 046.

[17] C. Ma, J. Zhang, Z. Zhu *et al.*, "AgentBoard: An analytical evaluation board of multi-turn LLM agents," in *Advances in Neural Information Processing Systems*, vol. 37. Neural Information Processing Systems Foundation, 2024, pp. 74 325–74 362.

[18] X. Zhou, H. Zhu, L. Mathur *et al.*, "SOTOPIA: Interactive evaluation for social intelligence in language agents," in *International Conference on Learning Representations*, 2024, pp. 40 975–41 019.

[19] D. Paglieri, B. Cupiał, S. Coward *et al.*, "BALROG: Benchmarking agentic LLM and VLM reasoning on games," in *International Conference on Learning Representations*, 2025, pp. 96 666–96 702.

[20] P. Le Pelletier de Woillemont, R. Labory, and V. Corruble, "Automated play-testing through RL based human-like play-styles generation," *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 18, no. 1, pp. 146–154, 2022.

[21] Y. Ma, J. Yan, B. Shi *et al.*, "Can LLM agents stick to the script? a benchmark for long-horizon consistency in interactive narratives," 2026, preprint (v1); ICML 2026 acceptance comment on arXiv, proceedings record unverified.

[22] S. Geng, M. Josifoski, M. Peyrard, and R. West, "Grammar-constrained decoding for structured NLP tasks without finetuning," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2023, pp. 10 932–10 952.

[23] L. Beurer-Kellner, M. Fischer, and M. Vechev, "Prompting is programming: A query language for large language models," *Proceedings of the ACM on Programming Languages*, vol. 7, no. PLDI, pp. 1946–1969, 2023.

[24] T. Scholak, N. Schucher, and D. Bahdanau, "PICARD: Parsing incrementally for constrained auto-regressive decoding from language models," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2021, pp. 9895–9901.

[25] S. Kambhampati, K. Valmeekam, L. Guan *et al.*, "Position: LLMs can't plan, but can help planning in LLM-Modulo frameworks," in *Proceedings of the 41st International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 235. PMLR, 2024, pp. 22 895–22 907.

[26] B. Ichter, A. Brohan, Y. Chebotar *et al.*, "Do as I can, not as I say: Grounding language in robotic affordances," in *Proceedings of the 6th Conference on Robot Learning (CoRL 2022)*, ser. Proceedings of Machine Learning Research, vol. 205. PMLR, 2023, pp. 287–318.

[27] H. Wang, C. M. Poskitt, and J. Sun, "AgentSpec: Customizable runtime enforcement for safe and reliable LLM agents," 2025, preprint (v3); ICSE 2026 acceptance stated on arXiv, proceedings record unverified.

[28] Z. Zeng, S. Li, J. Xi, A. Zhu, and P. Ammanabrolu, "Setting the DC: Tool-grounded D&D simulations to test LLM agents," in *NeurIPS 2025 Workshop on Generative and Protective AI for Content Creation (GenProCC)*, 2025, non-archival workshop paper; OpenReview/PDF list Zeng first, with Zeng and Li as equal contributors; NeurIPS/GenProCC lists display Li first.

[29] E. Zhou, S. Basavatia, M. Siam, Z. Chen, and M. O. Riedl, "STORY2GAME: Generating (almost) everything in an interactive fiction game," 2025, preprint.

[30] S. Buongiorno, L. Klinkert, Z. Zhuang, T. Chawla, and C. Clark, "PANGeA: Procedural artificial narrative using generative AI for turn-based, role-playing video games," *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment*, vol. 20, no. 1, pp. 156–166, 2024.

[31] M. Vaucher, S. Silveira, S. Góngora, and L. Chiruzzo, "IVIE: A neuro-symbolic approach to incremental and validated generation of interactive fiction worlds," in *Proceedings of the 17th International Conference on Computational Creativity (ICCC'26)*. Association for Computational Creativity, 2026, archival proceedings (ISBN); preprint arXiv:2606.13348.

[32] S. Góngora, L. Chiruzzo, G. Méndez, and P. Gervás, "World-state transformations for neuro-symbolic interactive storytelling," 2026, preprint.

[33] S. Geng, H. Cooper, M. Moskal *et al.*, "JSONSchemaBench: A rigorous benchmark of structured outputs for language models," 2025, preprint (v3).

[34] X. He, Y. Tian, Y. Sun *et al.*, "G-Retriever: Retrieval-augmented generation for textual graph understanding and question answering," in *Advances in Neural Information Processing Systems*, vol. 37. Neural Information Processing Systems Foundation, 2024, pp. 132 876–132 907.

[35] B. Jiménez Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su, "HippoRAG: Neurobiologically inspired long-term memory for large language models," in *Advances in Neural Information Processing Systems*, vol. 37. Neural Information Processing Systems Foundation, 2024, pp. 59 532–59 569.

[36] P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav, "Mem0: Building production-ready AI agents with scalable long-term memory," in *ECAI 2025*, ser. Frontiers in Artificial Intelligence and Applications, vol. 413. IOS Press, 2025, pp. 2993–3000.

[37] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," in *Advances in Neural Information Processing Systems*, vol. 34. Curran Associates, Inc., 2021, pp. 29 304–29 320.

[38] J. Pineau, P. Vincent-Lamarre, K. Sinha *et al.*, "Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.

[39] P. Henderson, J. Hu, J. Romoff, E. Brunskill, D. Jurafsky, and J. Pineau, "Towards the systematic reporting of the energy and carbon footprints of

machine learning," *Journal of Machine Learning Research*, vol. 21, no. 248, pp. 1–43, 2020.

[40] C. van der Lee, A. Gatt, E. van Miltenburg, S. Wubben, and E. Krahmer, "Best practices for the human evaluation of automatically generated text," in *Proceedings of the 12th International Conference on Natural Language Generation*. Association for Computational Linguistics, 2019, pp. 355–368.

[41] M. Karpinska, N. Akoury, and M. Iyyer, "The perils of using Mechanical Turk to evaluate open-ended text generation," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2021, pp. 1265–1285.

[42] L. Zheng, W.-L. Chiang, Y. Sheng *et al.*, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems*, vol. 36. Neural Information Processing Systems Foundation, 2023, pp. 46 595–46 623.

[43] G. H. Chen, S. Chen, Z. Liu, F. Jiang, and B. Wang, "Humans or LLMs as the judge? a study on judgement bias," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2024, pp. 8301–8327.

[44] D. Lakens, A. M. Scheel, and P. M. Isager, "Equivalence testing for psychological research: A tutorial," *Advances in Methods and Practices in Psychological Science*, vol. 1, no. 2, pp. 259–269, 2018.

[45] D. J. Barr, R. Levy, C. Scheepers, and H. J. Tily, "Random effects structure for confirmatory hypothesis testing: Keep it maximal," *Journal of Memory and Language*, vol. 68, no. 3, pp. 255–278, 2013.

[46] E. Feng, W. Zhou, Z. Liu *et al.*, "Get experience from practice: LLM agents with record & replay," 2025, preprint.

[47] Y. Sun, H. Wang, D. Li, G. Wang, and H. Zhang, "The emperor's new clothes in benchmarking? a rigorous examination of mitigation strategies for LLM benchmark data contamination," 2025, preprint (v1).

[48] A. Madaan, N. Tandon, P. Gupta *et al.*, "Self-Refine: Iterative refinement with self-feedback," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 46 534–46 594.

[49] M. O. Riedl, C. J. Saretto, and R. M. Young, "Managing interaction between users and agents in a multi-agent storytelling environment," in *Proceedings of the Second International Joint Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2003, pp. 741–748.

[50] R. E. Fikes and N. J. Nilsson, "STRIPS: A new approach to the application of theorem proving to problem solving," *Artificial Intelligence*, vol. 2, no. 3–4, pp. 189–208, 1971.

[51] F. B. Schneider, "Enforceable security policies," *ACM Transactions on Information and System Security*, vol. 3, no. 1, pp. 30–50, 2000.

[52] T. Haerder and A. Reuter, "Principles of transaction-oriented database recovery," *ACM Computing Surveys*, vol. 15, no. 4, pp. 287–317, 1983.

[53] R. Evans and E. Short, "Versu—a simulationist storytelling system," *IEEE Transactions on Computational Intelligence and AI in Games*, vol. 6, no. 2, pp. 113–130, 2014, predecessor title of IEEE Transactions on Games.

[54] S. G. Ware and C. Siler, "Sabre: A narrative planner supporting intention and deep theory of mind," in *Proceedings of the AAAI Conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, vol. 17, no. 1, 2021, pp. 99–106.

[55] N. Shinn, F. Cassano, A. Gopinath, K. Narasimhan, and S. Yao, "Reflexion: Language agents with verbal reinforcement learning," in *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023, pp. 8634–8652.

[56] X. Chen, M. Lin, N. Schärli, and D. Zhou, "Teaching large language models to self-debug," in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024, accepted poster; OpenReview archival record; preprint arXiv:2304.05128.

[57] A. Solar-Lezama, L. Tancau, R. Bodík, S. Seshia, and V. Saraswat, "Combinatorial sketching for finite programs," in *Proceedings of the 12th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*. ACM, 2006, pp. 404–415.

[58] Y. Huang, G. Chen, J. Yao *et al.*, "Beyond state consistency: Behavior consistency in text-based world models," 2026, preprint.

# Appendix A
# Runtime Evidence Appendix

## FIXED-GAME NATIVE RUNTIME COMPARISON I 8 ASSIGNED ROWS

One authored quest; n=2 per selector/representation cell; NR = not reported, not zero. Labels remain explicit; color is supplementary.

| Row | Requested selector | Representation / repeat | Status | Goal | Host calls | Elapsed (s) | Safety R record-link / E engine |
|---|---|---|---|---|---|---|---|
| 1 | gpt-6.1-sol | general_rule / r1 | HARNESS | NR | 0 | NR | NOT_OBSERVED R NR / E NOT_OBSERVED |
| 2 | gpt-6.1-sol | general_rule / r2 | SUCCESS | YES | 4 | 45.91 | PASS R YES / E VALID |
| 3 | gpt-6.1-sol | ontology / r1 | TIMEOUT | NO | 2 | 84.66 | PASS R YES / E VALID |
| 4 | gpt-6.1-sol | ontology / r2 | TIMEOUT | NR | 1 | 60.04 | NO_COMMIT_OBSERVED R YES / E NOT_RUN |
| 5 | gpt-5.6-sol | general_rule / r1 | SUCCESS | YES | 4 | 41.09 | PASS R YES / E VALID |
| 6 | gpt-5.6-sol | general_rule / r2 | SUCCESS | YES | 4 | 39.39 | PASS R YES / E VALID |
| 7 | gpt-5.6-sol | ontology / r1 | SUCCESS | YES | 4 | 36.68 | PASS R YES / E VALID |
| 8 | gpt-5.6-sol | ontology / r2 | SUCCESS | YES | 4 | 29.55 | PASS R YES / E VALID |

Endpoint outcomes, receipt-derived host calls, and dispatch-to-terminal elapsed time.

Typed-intent proposal; deterministic planner/checker and Godot commit retain authority.

No efficacy/superiority/noninferiority/player-benefit claim; provider metadata and request counts remain NR.

Fig. 7. Trace-derived native runtime comparison for the fixed-game pilot (one authored quest; 8 assigned rows, 7 observed executions). System-goal outcomes and assigned denominators retain unknown provider identity, revision, cost, and request-count fields; one pre-response harness-serializer technical failure and two frozen 60 s `gpt-6.1-sol` ontology response-timeout failures are distinguished from goal successes. Evidence image, not gameplay screenshot.

## APPENDIX B
## IDENTITY-REQUIRED EXTENSION

**Identity-admission stop: not a model comparison.** The separate extension has 20 planned rows, 1 assigned row, 1 native process, 0 identity-qualified rows and 19 unattempted rows. The first requested `gpt-6-luna`/`general_rule` CLI turn completed, but its response lacked provider model-identity evidence; it was not accepted and the batch stopped.

A requested selector or catalog advertisement is not a served identity. The evidence cannot distinguish an absent `openai-model` header from one not delivered as a valid string. This is not an observed model mismatch or model-incapability result. `gpt-6-terra` was not advertised by the pinned catalog, was not assigned and was not substituted.

Audit distinction: `record hash valid=true`; `engineering record accepted=false`. The original bootstrap label `episode_record_integrity_invalid` is retained. The recomputed hash matches; the compound predicate also requires engineering/safety PASS, so this label is not evidence of corrupted bytes. There were 0 commits, safety is `NO_COMMIT_OBSERVED`, and fresh-engine replay is `NOT_RUN`, not a safety pass.

Two blocked repair requests were not native reruns. The original eight-row pilot remains byte-identical and unpooled. Provider identity, revision, effective effort, request count, cost, actual credit use and cash charge remain `unknown`. Matching accounts and automatic reload OFF were observed before and after, without a setting change or purchase; these are not zero-cost or zero-credit attestations. No verified new model comparison, ontology effect or confirmatory A/C evidence follows.

Preparation 7.144 s; episode elapsed 4.257 s; native process 4.207 s; orchestration 11.566 s. These are distinct failed-attempt boundaries, not pure inference latency or time to success.

TABLE XV
TEN IDENTITY-ADMISSION CELLS DISTINGUISH PLANS, ASSIGNMENTS, QUALIFIED IDENTITY AND UNATTEMPTED ROWS; NOT MODEL PERFORMANCE.

| Requested selector | Representation | Planned | Assigned | ID-qualified | Not run |
|---|---|---|---|---|---|
| `gpt-6-luna` | `general_rule` | 2 | 1 | 0 | 1 |
| `gpt-6-luna` | `ontology` | 2 | 0 | 0 | 2 |
| `gpt-6-sol` | `general_rule` | 2 | 0 | 0 | 2 |
| `gpt-6-sol` | `ontology` | 2 | 0 | 0 | 2 |
| `gpt-5.6-luna` | `general_rule` | 2 | 0 | 0 | 2 |
| `gpt-5.6-luna` | `ontology` | 2 | 0 | 0 | 2 |
| `gpt-5.6-sol` | `general_rule` | 2 | 0 | 0 | 2 |
| `gpt-5.6-sol` | `ontology` | 2 | 0 | 0 | 2 |
| `gpt-5.6-terra` | `general_rule` | 2 | 0 | 0 | 2 |
| `gpt-5.6-terra` | `ontology` | 2 | 0 | 0 | 2 |
| Total | — | 20 | 1 | 0 | 19 |

# APPENDIX C
# LOCAL OLLAMA FEASIBILITY NOTE

This appendix records a bounded, descriptive feasibility exercise rather than a result or comparison study. Four single-episode fixed-game assignments were executed (4/4); all four failed to reach the game goal (0/4). The pilot row was executed before its plan-only receipt was written; the other three plan-only receipts preceded execution, so these rows are not a frozen or prospective cohort. The four rows remain the denominator, and response-qualified eligibility is reported separately from success. The requested and response model names agree on the 21 non-timeout responses. The two timeout receipts have no response-model observation; in particular, the qwen2.5:0.5b `general_rule` run has no response-model observation. These receipts are not treated as identity evidence. Ollama was version 0.30.10, and the model manifest/blob digests matched before and after the runs. Provider identity, revision, cost, hardware, load, and reproducible timing are unavailable. No network-observation claim is made beyond the configured loopback endpoint.

The two representations are serializations of the same public facts, not independent semantic conditions. The language model classified the public utterance; a symbolic planner selected the action. These episodes therefore do not test an ontology effect, model-size effect, ranking, planning ability, efficacy, usability, or player benefit. The failure taxonomy is descriptive: one host-call-budget exhaustion, two response-timeout events under the cold-load deadline, and one repair-source error with four retained malformed-response receipts. Raw malformed responses were not retained, so their causes cannot be diagnosed.

Rows 1–3 had no state-changing commit and retained the initial state hash; row 4 had one state-changing commit and replayed validly in a fresh Godot process. Replay used zero model calls. The record, receipt, and request hashes verify, but the adapter and runner source bytes were not contemporaneously hash-bound to the records; these artifacts are therefore not fully reproducible. Absolute local paths remain only in the local records, which are not released.

Local-only packets retain the pilot episode, verification and post-hoc plan-only receipt, alongside the continuation records for the other three episodes. They support the bounded observations above, but do not resolve the source-binding limitation.

TABLE XVI
LOCAL OLLAMA FIXED-GAME FEASIBILITY OBSERVATIONS. EACH ROW IS ONE ASSIGNED EPISODE; RECEIPT COUNTS ARE VERIFIED/FAILED/TIMEOUT. ELIGIBILITY MEANS THAT EVERY RECEIPT WAS RESPONSE-QUALIFIED, NOT THAT THE GOAL WAS REACHED.

| Model | Form | Assigned/executed | Receipts | Failure code | Eligible | State/replay | Engineering status |
|---|---|---|---|---|---|---|---|
| qwen2.5:1.5b | `general_rule` (pilot) | 1/1 | 12/0/0 | BUDGET | yes | no-change / valid | NOT_PASS |
| qwen2.5:1.5b | `ontology` | 1/1 | 4/0/1 | TIMEOUT | no | no-change / valid | NOT_PASS |
| qwen2.5:0.5b | `general_rule` | 1/1 | 0/0/1 | TIMEOUT | no | no-change / NOT_RUN | NOT_PASS |
| qwen2.5:0.5b | `ontology` | 1/1 | 1/4/0 | REPAIR | no | 1 change / valid | PASS |

Failure codes: BUDGET = `host_call_budget_exhausted`; TIMEOUT = `response_timeout`; REPAIR = `repair_source_error`. The 4-row roll-up is assigned/executed $4/4$, goal reached $0/4$, and response-qualified eligibility $1/4$. The one engineering PASS records a valid repaired state-changing step; it is not task success. The 30-second response deadline included cold model loading with `keep_alive=0`.